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THE NATIONAL COLLEGE BASAVANAGUDI, BENGALURU- 560 004

AUTONOMOUS NAAC Accredited 'A' Grade

Affiliated to the Bengaluru City University Website: www.ncbgudi.com

Curriculum Structure

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B.Sc in Computer Science



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Syllabus of Third and Fourth Semesters for

B.Sc. Computer Science (CBCS Scheme) Under State Education Policy

**Effective from the Academic Year
2025 – 2026**

Name of the Degree Program	B.Sc. Computer Science
Discipline Course	Computer Science
Starting Year of Implementation	2024-25 (I & II Semesters)
	2025-26 (III & IV Semesters)
	2026-27 (V & VI Semesters)

Programme Outcomes (POs)

PO 1	Apply knowledge of computer science to design and implement computational solutions.
PO 2	Identify, analyze, and solve structured and unstructured real-world problems using algorithmic thinking, data structures and efficient programming strategies.
PO 3	Create, Solve and apply tools of programming, networking, database and modern technology for solutions to significant problems.
PO 4	Design computer-based solutions for various technical problems.
PO 5	Demonstrate project management skills through projects and practical labs.
PO 6	Build software solutions using multiple paradigms in software development.
PO 7	Understand professional, ethical, legal and societal responsibilities in data usage, privacy, AI implementations and technological impact.
PO 8	Communicate effectively in both verbal and written formats through project reports, lab records presentations and document code.
PO 9	Function effectively as an individual and as a member or leader in diverse teams in multidisciplinary environments such as group projects and internships.
PO 10	Recognize the need for continuous learning in a rapidly evolving tech world and demonstrate readiness to adopt innovations in AI for solving real-world challenges.

ASSESSMENT

Weightage for the Assessments (in percentage)

Type of Course	Formative Assessment/ I.A.	Summative Assessment (S.A.)
Theory	20%	80%
Practical	20%	80%

B.Sc. (Computer Science as an Optional)
AS PER STATE EDUCATION POLICY
DETAILED STRUCTURE FOR B.Sc. COMPUTER SCIENCE

S e m e s t e r	Course Code	Paper Title	Teac hing Hou rs / Wee k	Marks		T ot al M ar ks	Dur atio n of Exa m in Hou rs	C r e d i t s
				Exam	IA			
I	24BSC-CS-1	Problem Solving Techniques	04	80	20	100	03	03
	24BSC-CS-1P	Problem Solving Techniques Lab	04	40	10	50	03	02
II	24BSC-CS-2	Data Structures	04	80	20	100	03	03
	24BSC-CS-2P	Data Structures Lab	04	40	10	50	03	02
III	24BSC-CS-3	Database Management System	04	80	20	100	03	03
	24BSC-CS-3P	Database Management System Lab	04	40	10	50	03	02
	24BSC-CS-E1	Cyber Security	02	40	10	50	1.5	02
IV	24BSC-CS-4	Object Oriented Programming using Java	04	80	20	100	03	03
	24BSC-CS-4P	JAVA Lab	04	40	10	50	03	02
	24BSC-CS-E2	Software Testing	02	40	10	50	1.5	02
	24BSC-CS-SE1	HTML	02	40	10	50	1.5	02

B.Sc. (Computer Science as an Optional)
AS PER STATE EDUCATION POLICY
DETAILED STRUCTURE FOR B.Sc. COMPUTER SCIENCE

S e m e s t e r	Course Code	Paper Title	Teac hing Hou rs / Wee k	Marks		T ot al M ar ks	Durat ion of Exam in Hour s	C r e d i t s
				Exam	IA			
V	24BSC-CS-5	Python Programming	04	80	20	100	03	03
	24BSC-CS-6	Operating System	04	80	20	100	03	03
	24BSC-CS-5P	Python Programming Lab	04	40	10	50	03	02
	24BSC-CS-6P	Operating System Lab	04	40	10	50	03	02
	24BSC-CS-SE2	Computer Graphics	02	40	10	50	1.5	02
VI	24BSC-CS-7	Artificial Intelligence	04	80	20	100	03	03
	24BSC-CS-8	Computer Networks	04	80	20	100	03	03
	24BSC-CS-7P	Artificial Intelligence Lab	04	40	10	50	03	02
	24BSC-CS-8P	Computer Networks Lab	04	40	10	50	03	02

BSC Course Matrix – SEP -2025

B.Sc. -- III -- Semester Year 2025-26								
	Subjects			Marks				
Code	Course	L	P	Exam	IA	Total	Exam Duration	Credits
24BSC-CS-3	Database Management System	04	-	80	20	100	03	03
24BSC-CS-3P	Database Management System Lab	-	04	40	10	50	03	02
24BSC-CS-E1	Cyber Security	02	-	40	10	50	1.5	02

B.Sc. – IV -- Semester Year 2025-26								
	Subjects			Marks				
Code	Course	L	P	Exam	IA	Total	Exam Duration	Credits
24BSC-CS-4	Object Oriented Programming using Java	04	-	80	20	100	03	03
24BSC-CS-4P	JAVA Lab	-	04	40	10	50	03	02
24BSC-CS-E2	Software Testing	02	-	40	10	50	1.5	02
24BSC-CS-SE1	HTML	02	-	40	10	50	1.5	02

B.Sc. 3rd Semester Syllabus 2025 -2026

TITLE: DATABASE MANAGEMENT SYSTEM			
B.Sc. – III -- Semester			
Paper Code -- 24BSC-CS-3		Theory -- Paper	
Teaching Hours : 04 Hours/Week		Duration of Exam : 03 Hours	
Lecture Hrs. 56	Internal Marks : 20	Exam Marks : 80	Credits: 3
Course Outcomes (COs) ➤ CO1: Understand the basic concepts of DBMS and its types, data independence and three schema architecture. ➤ CO2: Be familiar with the CODD's rules, E-R Model and the structure of the Relational databases ➤ CO3: Understand the concept of normalization and different types of normalization. Design normalized database objects and process the data in an optimized way ➤ CO4: Understand the basics of query evaluation techniques and query optimization and transaction processing.			

Unit- I	Introduction to Databases and Architecture	14 Hours
Introduction to Data and Database. Significance of Database Management System, Types of DBMS, Data Independence, Three Levels of Architecture, Client/Server Architecture, System Structure, Instance and Schema.		
Unit- II	Data Modeling and SQL Basics	14 Hours
Keys, CODD's Rules, Design Issues, ER – Model, Attribute types, Weak Entity Sets, Extended ER Features, ER to Relational Mapping, Structure of Relational Databases, Creation and Manipulation of Database using SQL commands (DDL, DML,DCL,TCL).		
Unit- III	Normalization and Functional Dependencies	14 Hours

Functional Dependency, Anomalies, Armstrong's axioms: Closure of a relation and Closure of attribute, Normalization, Types of normal forms (:1NF, 2NF, 3NF, Boyce - Codd Normal Form), Lossless decomposition.

Unit- IV	Relational Algebra and Transaction Management	14 Hours
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The Relational Algebra, Query Processing and Optimization, Evaluation of Relational algebraic expressions, Query Equivalence, Transaction Processing: ACID properties, states of a transaction, Introduction to concurrency control, Deadlock-Recovery. Introduction to PL/SQL.

TEXT BOOKS:

1. Elmasri and Navathe: Fundamentals of Database Systems, 7th Edition, Addison – Wesley, 2016.
2. Silberschatz, Korth and Sudharshan Data base System Concepts, 7th Edition, Tata McGraw Hill, 2019.

REFERENCE BOOKS:

1. C.J. Date, A. Kannan, S. Swamynatham: An Introduction to Database Systems, 8th Edition, Pearson education, 2009
2. Database Management Systems :Raghu Ramakrishnan and Johannes Gehrke: 3rd Edition, McGraw-Hill, 2003

TITLE : DATABASE MANAGEMENT SYSTEM LAB	
B.Sc. – III -- Semester	
Paper Code : 24BSC-CS-3P	Hands On - Lab
Lab Teaching Hours : 03 Hours/Week	Duration of Exam : 03 Hours
Maximum Marks : 50 (Exam 40 + IA 10)	Credits : 3
PART-A	
1.	Create a table STUDENT with the following fields: RollNo, Name, DOB, Department, Marks.
	➤ Insert at least 5 records. ➤ Display all records. ➤ Update marks for a specific student. ➤ Delete one student record.
2.	Create a table COURSE with CourseID, CourseName, and Credits.
	➤ Alter the table to add a field Department. ➤ Drop the field Credits.
3.	Create two tables EMPLOYEE1 and EMPLOYEE2.
	➤ Perform UNION, INTERSECT and MINUS operations on them.
4.	Use the STUDENT table to perform the following.
	➤ Find average marks. ➤ Display names in uppercase. ➤ Extract the month from DOB.
5.	Using the STUDENT and COURSE tables.

	➤ Find students who scored above average marks using a sub query. ➤ List students enrolled in 'Computer Science' using a sub query.
6.	Create DEPARTMENT and STUDENT tables.
	➤ Write queries to display student names along with their department names using ➤ INNERJOIN and LEFT JOIN.
7.	Create a view to show student names and marks from the STUDENT table where marks > 75.
	➤ Query the view ➤ Try updating the view
8.	Demonstrate the use of GRANT and REVOKE on the STUDENT table.
	➤ Use COMMIT and ROLLBACK after INSERT and DELETE commands.

TITLE : DATABASE MANAGEMENT SYSTEM LAB	
B.Sc. – III -- Semester	
Paper Code : 24BSC-CS-3P	Hands On - Lab
Lab Teaching Hours : 03 Hours/Week	Duration of Exam : 03 Hours
Maximum Marks : 50 (Exam 40 + IA 10)	Credits : 3
PART-B : PL/SQL PROGRAMMING	
9.	Write a PL/SQL block to accept a number and check if it is even or odd.
10.	Write a PL/SQL program to divide two numbers and handle the exception if the denominator is zero.
11.	Create an explicit cursor to display all student records with marks greater than 80.

12.	Use a parameterized cursor to display students based on department input.
13.	Create a stored function to calculate grade based on marks:
	➤ Above 80: Distinction ➤ 60–79: First Class ➤ 40–59: Second Class ➤ Below 40: Fail
14.	Write a stored procedure to update the marks of a student given their RollNo and marks.

TITLE : CYBER SECURITY		
B.Sc. – III -- Semester		
Paper Code : 24BSC-CS-E1	Theory : Elective - 1	
Teaching Hours : 02 Hours/Week	Duration of Exam 1:30 Hours	
Maximum Marks : 50 (Exam 40 + IA 10)	Credits : 2	
Lecture Hrs. 28	Internal Marks : 10	Exam Marks : 40
Course Outcomes: ➤ CO1 : Understand the fundamentals of cyber security and adopt safe internet practices and responsible digital behavior, and concepts of cryptography, Password security and secure communication protocols.		

- **CO2 :** Apply system and personal security measures to protect devices and data From threats, vulnerabilities and online attacks.

UNIT - I	Basics of Cyber Security and Cryptography	14 Hours
Introduction to cyber security, Importance of cyber safety in the digital world, Types of cyber threats – viruses, worms, Trojans, phishing, ransom ware, fake websites, social engineering, Goals of cyber security – confidentiality, integrity, availability, Difference between threats, vulnerabilities and risks, Common security terms – firewall, antivirus, malware, hacking, Overview of safe internet practices and responsible digital behavior. What is cryptography – Purpose of encryption and decryption, Difference between secret key (symmetric) and public key (asymmetric) encryption, Simple examples – Caesar cipher, concept of public/private keys, Introduction to passwords and password policies, Importance of HTTPS and SSL in secure communication,		
UNIT - II	System and Personal Security Practices	14 Hours
Basics of operating system security, User authentication and access control, Password policies and best practices, Role of antivirus and anti-malware software, Common web security issues – SQL injection, XSS, CSRF (conceptual overview), Mobile app security – permissions and safe usage, Importance of software updates and patch management. Safe use of mobile devices, laptops, and social media, Secure browsing practices and identifying fake or unsafe websites, Safe use of public Wi-Fi and avoiding data leaks, Risks of pop-ups, cookies, and online tracking, Introduction to firewalls and VPNs (concept only), Email safety – detecting phishing and spam, Responsible online behavior and data sharing.		

TEXT BOOKS:

1. Chirag Shah (2018). A Hands-On Introduction to Cyber security. Wiley.
2. Nina Godbole & Sunit Belapure (2011). Cyber Security: Understanding
3. Cybe Crimes, Computer Forensics and Legal Perspectives. Wiley India.
4. Rao, U. & Nayak, P. (2014). Cyber Security. Cengage Learning.

5. Moeti J. (2021). Cyber security for Beginners. Amazon Digital Services (for simple reading-level coverage).

B.Sc. 4th Semester Syllabus 2025 -

TITLE: OBJECT ORIENTED PROGRAMMING USING JAVA			
B.Sc. – IV -- Semester			
Paper Code -- 24BSC-CS-4		Theory -- Paper	
Teaching Hours : 04 Hours/Week		Duration of Exam : 03 Hours	
Lecture Hrs. 56	Internal Marks : 20	Exam Marks : 80	Credits: 3
Course Outcomes			
➤ CO1 Understand OOPs concepts. ➤ CO2 Analyze Polymorphism and different forms of Inheritance and packages ➤ CO3 Understand different GUI layouts and design GUI applications ➤ CO4 Understand different kinds of file I/O, Multithreading and Exception handling			
UNIT- I Introduction to Java and Oop's concepts 14 Hours			
Basics of Java programming, Data types, Variables, Operators, Control structures including selection, Looping, Java methods, Overloading, Math class, Arrays in java. Objects and Classes: Basics of objects and classes in java, Constructors, Finalize, Visibility modifiers, Methods and objects, Inbuilt classes like String, Character, String Buffer, File, this reference			
UNIT- II Inheritance and Packages 14 Hours			

Inheritance in java, Super and sub class, Overriding, Object class, Polymorphism, Dynamic binding, Generic programming, Casting objects, Instance of operator, Abstract class, Interface in java, Package in java, UTIL package.		
UNIT- III	Event Handling and GUI Programming	14 Hours
Event handling in java, Event types, Mouse and key events, GUI Basics, Panels, Frames, Layout Managers: Flow Layout, Border Layout, Grid Layout, GUI components : Buttons, Check Boxes, Radio Buttons, Labels, Text Fields, Text Areas, Combo Boxes, Lists, Scroll Bars, Sliders, Windows, Menus, Dialog Box, Applet and its life cycle, Introduction to swing.		
UNIT- IV	Exception handling, Multithreading and I/O Programming	14 Hours
Exception handling mechanism, try-catch-finally, Thread life cycle and methods, Runnable interface, Thread synchronization. I/O programming: Text and Binary I/O, Binary I/O classes, Object I/O, Random Access Files.		

TEXT BOOKS

1. E. Balagurusamy, Programming with JAVA, McGraw Hill, New Delhi, 2007
- 2.

REFERENCE BOOKS

3. Herbert Schildt, Java A Beginner's Guide – Create, Compile, and Run Java Programs Today, Sixth Edition, Oracle Press, 2014
4. Ken Arnold, James Gosling, “The Java Programming Language, Fourth Edition, Addison Wisely, 2005
5. Herbert Schildt, ‘The Complete Reference Java, 7th Edition, McGraw Hill, 2007

TITLE : JAVA LAB	
B.Sc. – IV -- Semester	
Paper Code : 24BSC-CS-4P	Hands on Lab
Lab Teaching Hours : 03 Hours/Week	Duration of Exam : 03 Hours
Maximum Marks : 50 (Exam 40 + IA 10)	Credits : 3
PART - A	
1.	Write a Java Program to implement the usage of static, local and global variables.
2.	Write a Java Program to implement default and parameterized constructors.
3.	Write a Java Program to implement java String functions.
4.	Write a Java Program to implement array of objects.
5.	Write a Java Program to implement multiple inheritance using interface.
6.	Write a Java Program to implement run time polymorphism
7.	Write a Java Program to catch negative array size exception

8.	Write a Java Program to add two integers and two float numbers. When no arguments are supplied, give a default value to calculate the sum. Use function overloading.
9.	Create a simple applet which reveals your personal information.
10.	Write a Java Program to create a thread using Runnable interface.
PART - B	
11.	To find the area and circumference of the circle by accepting the radius from the user
12.	Write a Java Program to find maximum of three numbers.
13.	Write a Java Program to list the factorial of numbers from 1 to 10.
14.	Write a Java Program to implement single inheritance
15.	Write a Java Program to import user defined package .
16.	Write a Java Program to demonstrate division by zero exception

TITLE : SOFTWARE TESTING		
B.Sc. – IV -- Semester		
Paper Code : 24BSC-CS-E2	Theory : Elective - 2	
Teaching Hours : 02 Hours/Week	Duration of Exam : 1:30 Hours	
Maximum Marks : 50 (Exam 40 + IA 10)	Credits : 2	
Lecture Hrs. 28	Internal Marks : 10	Exam Marks : 40
Course Outcomes ➤ CO1 Understand software development life cycle, models and various testing concepts. ➤ CO2 Understand defect management to improve software quality and reliability. Learning Agile to collaborate effectively in modern development environments.		
UNIT - I	Software Development and Testing	14 Hour
Software Development Life Cycle, Waterfall model, Prototype model, Spiral model. Manual Software Testing Concepts: Introduction to Manual Testing, Types Of Software Testing: Unit Testing, Integration Testing, System Testing,		

And Acceptance Testing. White Box Testing, Grey Box Testing, And Black Box Testing. Software Testing Life Cycle (STLC).		
UNIT - II	Defect and Agile Methodology	14 Hour
Blocker defect, Critical defect, Major defect. Defect Life Cycle, Agile methodology: definition, principles and model, Scrum: Sprint Planning, Meeting, Scrum Master, Scrum meeting. Automation Testing: Introduction to Selenium.		
TEXT BOOKS 1. Dorothy Graham, Rex Black, Erik van veenendaal, Foundations of Software Testing(Paperback), 4 edition, Cengage Learning India Pvt. Ltd., 2020 2. Rahul Verma, The Last Book on Testing Paperback, 1 edition, India, 2025		

TITLE : HTML		
B.Sc. – IV -- Semester		
Paper Code : 24BSC-CS-SE1	Theory: Skill Enhancement Course-1	
Maximum Marks : 50 (Exam 40 + IA 10)	Credits : 2	
Lecture Hrs. 28	Internal Marks : 10	Exam Marks : 40
Course Outcomes ➤ CO1 Understand WWW, Importance of HTML, Basic tags ➤ CO2 Work with Lists ,Tables, forms and Practice Hyper linking.		
UNIT - I	Introduction to WWW and HTML tags	14 Hours
Basic principles involved in developing a web site, Brief History of Internet, introduction to World Wide Web and Web site, Web Standards, HTML: definition, significance, applications and limitations. Basic structure of an HTML document, Creating an HTML document, Basic Formatting Tags. Mark up Tags, Headings, Paragraphs, Line Breaks, Grouping Using Div and Span, Comments, HTML Code		

Example.

UNIT - II	Elements of HTML	14 Hours
Introduction to elements of HTML, Working with Text, Lists : Tags, significance of lists, types of Lists, creating lists, Tables: table tags, Creating tables, Forms and Controls: Importance of forms in websites, Form syntax ,creating html forms, Hyperlink: Working with Hyperlinks, Images and Multimedia, anchor tag, the importance of hyperlinks in web pages. HTML Meta tag, XHTML.		
TEXT BOOKS: 1. Kogent Learning Solutions Inc., web technologies HTML, javascript, PHP, Dreamtech Press, 1 Edition,2009 2. Steven M. Schafer,HTML, XHTML, and CSS Bible(paper back), 3 Edition, John Wiley & Sons, 004		

BLUEPRINT FOR QUESTION PAPER						
FOR 03 CREDITS						
Mark s	Unit I	Unit II	Unit III	Unit IV	Number of questions to be answered	Total
2	3	3	3	3	10	20
6	2	2	2	2	5	30
10	1	1	1	1	1	30
Total						80

BLUEPRINT FOR QUESTION PAPER				
FOR 02 CREDITS				
Marks	Unit I	Unit II	Number of questions to be answered	Total
2	3	3	5	10

5	3	3	4	20
10	1	1	1	10
Total				40

Formative Assessment - 03 Credits	
Category	Marks Allotted
Test	10
Assignment	10
Total Marks	20

Formative Assessment - 02 Credits	
Category	Marks Allotted
Test	05
Assignment	05
Total Marks	10